Altwiesenstrasse 89, Zurich
+41 76 639 5586
+86 13625283399

Yucheng Wang

yuchwang@ethz.ch
deltawang7@gmail.com
ycwang7@foxmail.com

Personal Website · Google Scholar

Education

ETH Zurich **Sept. 2024 - Present**

- **Master of Science in Computer Science**, major in Machine Intelligence
- Expected graduation: February 2027. Research interests: audio and multimodal large language models, understanding and reasoning, and multimodal agents.

Joint Bachelor's Degree Program **Sept. 2019 - Jun. 2024**

Columbia University **New York, NY**

- **Bachelor of Arts in Computer Science (Summa Cum Laude)**, GPA: 4.09/4.00 (Top 5%)
- Relevant Coursework: System Programming, Computational Linear Algebra, Computer Science Theory, Artificial Intelligence, Machine Learning, Deep Learning, Applied Computer Vision, High-performance Machine Learning, Computational Aspects of Robotics, Machine Learning with Applications in Finance

City University of Hong Kong **Kowloon, Hong Kong**

- **Bachelor of Science in Computer Science (First Class Honors)**, GPA: 4.01/4.30 (Top 5%)
- Relevant Coursework: Computer Programming, Calculus, Probability and Statistics, Discrete Mathematics, Computer Organization, Computer Networks, Data Structures, Operating Systems, Software Engineering, Database System, Design and Analysis of Algorithms, Multimodal Interface Design

Work Experience

Tencent AI Lab **Shenzhen, China** **Jun. 2025 - Sept. 2025**

Research Intern

- Contributed to music encoder pre-training for general-purpose large audio models; built training/evaluation pipelines and reproduced MuQ-based self-supervised pre-training.
- Combined self-supervised representation learning with music captioning and Contrastive Language-Audio Pretraining (CLAP) objectives to investigate joint training for richer music representations.
- Designed a fine-grained contrastive learning objective using frame-level chord pseudo-labels to preserve local harmonic structure that global audio-text alignment can overlook. Ablation studies showed gains of 31.2 and 26.4 percentage points in downstream chord and key recognition accuracy, respectively, while maintaining performance on other music understanding tasks.
- Associated manuscript submitted to ICASSP 2027.

Aispeech **Suzhou, China** **Jun. 2024 - Sept. 2024**

Machine Learning Engineer Intern, *Cloud Speech Recognition Group*

- Led the training, optimization, and delivery of a speech emotion recognition model for Mercedes-Benz's vehicle dialogue system, improving audio emotion detection accuracy by 13 percentage points over the previous system on the acceptance test set.
- Designed and implemented a multi-modal fusion algorithm to synergize outputs from speech and text emotion recognition models, achieving superior accuracy in real-time emotional state classification and aligning closely with client demands for realistic product integration.
- Trained a speech recognition model for Thai language using Feedforward Sequential Memory Networks (FSMN), enhancing speech processing capabilities and model adaptability to local dialects.
- Co-first authored a comprehensive survey paper on Speech Large Language Models (LLMs), published in **IEEE JSTSP 2025** and available on arXiv: Jing Peng*, **Yucheng Wang***, Yu Xi, Xv Li, and Kai Yu. **A Survey on Speech Large Language Models for Understanding.**

iFlytek **Suzhou, China** **Jun. 2022 - Aug. 2022**

Machine Learning Engineer Intern, *Computer Vision Group*

- Collaborated in the development of a Parkinson's disease identification system using human pose estimation techniques.
- Engineered a pipeline to filter consistent human body key points from depth and RGB camera images, preparing data for semi-supervised learning and significantly reducing the workload of manual labeling.
- Independently developed a Human Keypoints Labeling software using PyQt, enhancing manual data annotation efficiency.

Research Experience

Audio-Mind **Submitted to ICASSP 2027**

Co-first Author, *Collaboration with X-LANCE Lab, Shanghai Jiao Tong University*

- **Yucheng Wang***, Jing Peng*, Hanqi Li, Chenghao Wang, *et al.* **Audio-Mind: An Auditable Agentic Framework for Audio Understanding.**
- Led the engineering architecture and benchmark evaluation of a training-free, modular audio reasoning agent. A large audio-language model produces question-directed audio observations, while a central text LLM plans tool calls and targeted re-listening to acquire additional evidence as needed.
- Implemented an auditable agent workflow covering perception, planning, tool execution, evidence fusion, and answer verification, with explicit records of perceptual outputs, planning decisions, tool evidence, and uncertainty for error analysis.
- Achieved 80.4% accuracy on MMAR and 82.8% on MSU-Bench; gains were concentrated on difficult questions requiring multiple steps of evidence acquisition.

Audio Evidence Localization **Master's Thesis**

Evidence Localization and Targeted Re-listening in End-to-End Large Audio Models

- Proposed incorporating input-side audio tokens into the output sequence as retrievable evidence units, enabling models to explicitly extract relevant audio segments before answering questions and reason from localized evidence.
- Built 530K audio evidence retrieval pre-training examples and 30K high-quality evidence-grounded question-answering examples. Designed staged fine-tuning to first learn evidence extraction and segment localization, then strengthen question answering based on the extracted evidence.
- Conducted initial fine-tuning experiments with Qwen2.5-Omni and ablation studies demonstrating the feasibility of explicit audio evidence localization and the value of evidence outputs for subsequent audio reasoning.

Compose and Fuse **ICLR 2026**

Co-first Author

- **Yucheng Wang***, Yifan Hou*, Aydin Javadov, Mubashara Akhtar, Mrinmaya Sachan. **Compose and Fuse: Revisiting the Foundational Bottlenecks in Multimodal Reasoning.**
- Built a controlled diagnostic framework based on propositional logic, rendering facts as text, images, and audio across six cross-modal interaction patterns to examine when additional modalities help, interfere with, or fail to affect reasoning.
- Identified composition and fusion bottlenecks and modality bias in multimodal reasoning, showing that information integration is a key limitation beyond perception alone.
- Designed two-stage reasoning prompts and early-layer attention softening through temperature intervention to mitigate cross-modal evidence integration failures.

Speech LLM Survey **IEEE JSTSP 2025**

Co-first Author

- Jing Peng*, **Yucheng Wang***, Yu Xi, Xv Li, and Kai Yu. **A Survey on Speech Large Language Models for Understanding.**
- Systematically reviewed speech LLMs for understanding tasks and developed a taxonomy spanning speech information types, task objectives, and input-output formats.
- Synthesized model architectures, training strategies, datasets, and evaluation methods; analyzed challenges including instruction sensitivity and degradation of semantic reasoning capabilities to inform subsequent research on audio understanding and reasoning.

G-STAR **Findings of EMNLP 2026**

- Jing Peng, Ziyi Chen, Haoyu Li, **Yucheng Wang**, *et al.* **G-STAR: End-to-End Global Speaker-Tracking Attributed Recognition.**

Earlier Research Experience

City University of Hong Kong **Kowloon, Hong Kong** **Jun. 2021 - Apr. 2023**

Research on Human-Computer Interaction, *Supervisor: Dr. Zhicong LU, Assistant Professor at the Department of Computer Science, City University of Hong Kong*

- Utilized sophisticated computational models to analyze user behavior patterns in online communities, uncovering insights into the relationship between real-life sports events and online discussions, contributing to online community design.
- Developed an innovative metric to measure the gap between pre-match expectations and actual match results, addressing the issue of low-scoring football matches not accurately reflecting fans' impressions, thereby quantifying previously elusive aspects of fan sentiment.
- Built a visualization system for a prototype online sports discussion forum, integrating findings from user behavior pattern analysis. Implemented using front-end JavaScript frameworks such as Bootstrap, Vue.js, and Chart.js.
- Published the paper based on the research result which is accepted by CHI'23: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems: **Yucheng Wang**, Zhicong Lu. **Making Sense of Post-match Fan Behaviors in Football Communities.**

CityU Shenzhen Research Institute **Shenzhen, China** **Aug. 2021 - Mar. 2022**

Research Assistant, *Supervisor: Dr. Zhenjiang LI, Associate Professor at the Department of Computer Science, City University of Hong Kong*

- Collaborated on a research project focusing on deep neural network optimization for MCUs (Microcontroller units), applying pointwise pruning techniques to convolutional neural networks.
- Conducted experiments on the research by coding and deploying TinyML applications on microcontrollers and mobile devices.

Honors and Awards

- **Member of Phi Beta Kappa Society** for graduates in the 2023-2024 academic year - Inducted for demonstrating exceptional academic integrity and achievement in the liberal arts and sciences at Columbia University.
- **Russell C. Mills Award** (2023-2024) - Cash prize awarded by Columbia University's computer science department to a major demonstrating outstanding excellence in computer science.
- **George B. Bernheim Scholarship** for the 2023-2024 academic year - Named scholarship awarded by Columbia University (USD 40,000).
- **Dean's list** at Columbia University School of General Studies (all semesters, including Fall 2022 and Spring 2023).
- **Outstanding Academic Papers by Students** at City University of Hong Kong 2022 - Awarded recognition for my Final Year Project, "Online Community User Behavior Pattern Analysis," at City University of Hong Kong, 2022 (Granted to three students annually within the Department of Computer Science).
- **Bronze Award in the Global AI Challenge for Building E&M Facilities** 2021-2022 - Proposed a solution based on gradient boosting decision tree algorithm for forecasting building cooling load.
- **YEUNG Kin Man Scholarships** for CityU-Columbia U Joint Bachelor's Degree Program 2021-22 (HKD 150,000).
- **Dean's list** of College of Engineering at City University of Hong Kong (Spring 2020, Fall 2020, Spring 2021, Spring 2022).

Memberships

- Member of the Honor Society of the Columbia School of General Studies.
- Member of the Columbia Chapter of Upsilon Pi Epsilon, the International Computer Science Honor Society.
- Member of The ACM Special Interest Group on Computer-Human Interaction (SIGCHI).
- Member of CityU Golden Key Club.

Skills and Interests

- Programming languages: C/C++, Python, Java, SQL, HTML/CSS/JavaScript
- Tools and Frameworks: Git, Linux, Make, PyTorch, TensorFlow, LangGraph, CUDA, LaTeX
- Languages: Chinese (Native), English (Proficient; TOEFL 109, IELTS 7.5)
- Interests: Basketball (Member of Mainland Basketball Team of City University of Hong Kong), Music Production, Violin